

⇒ Incident Response Report:→ Simulated Ransomware Attack:→ Incident Identification:

- Incident Type: Ransomware Attack.
- Date: 03 August 2026
- Affected System: Windows files servers (FS-01)
- Detected by: Endpoint Detection and Response (EDR) Alert.

→ Severity: Critical.→ Collection phase:

- Windows Event logs
- Firewall logs.
- Antivirus / EDR logs.
- Ransomware note
- Encrypted files.
- Memory Dump
- Network traffic (PCAP)
- User login logs.

⇒ Preservation phase:

- The infected server was isolated from the network.
- A forensic disk image was created.
- Memory dump was collected before shutdown.
- SHA-256 hash values were generated for all devices

→ Original evidence was stored securely, and analysis was performed only on copies

⇒ Analysis phase:

- Initial access occurred through a phishing email attachment.
- malware executed after the user opened the attachment.
- The ransomware spread through shared network folders.

→ Sensitive company documents were encrypted.

→ Backup server remained unaffected.

⇒ Reporting phase:

→ Impact:

- Business operations disrupted.
- Important files encrypted.

⇒ Recommendations:

- Restore files from clean backups.
- Antivirus update & EDR signatures.

⇒ Evidence documentation log for a phishing incident:

Evidence ID	Evidence collected	Date & Time	collected By	Integrity Preserved.
E001	Phishing Email	03-08-2026 10:15AM	Security Analyst	SHA-256 hash generated
E002	Email header	03-08-2026 10:17AM	Security Analyst	Stored Read-only
E003	Suspicious URL	03-08-2026 10:20AM	SOC Analyst	Hash records & Screenshots
E004	Browser History	03-08-2026 10:30AM	Incident Responder	Forensic copies Created
E005	System logs	03-8-2026 10:40AM	SOC Team	Exported without modification
E006	Antivirus logs	03-08-2026 10:45AM	SOC Team	Digital Signature Verified
E007	Memory Dump	03-08-2026 10:55AM	Forensic investigators	SHA256 Verified.

⇒ Chain of custody:

Date	Person	Action
03-08-2026	Security Analyst	Evidence collected
03-08-2026	forensic investigators	Evidence collected
03-08-2026	incident Response Team	evidence collected.

⇒ Integrity preservation methods:

- SHA - 256 hashing.
- Read-only storage.
- Write blockers
- Secure evidence repository.
- Chain of custody documentation.

⇒ Dark web monitoring Alert:

The company's Dark web monitoring system generated an alert indicating that employee login credentials has been discovered on a dark web market place.

⇒ Incident identification:

- Incident Type: Credential leak.
- Source: Dark web monitoring Alert.
- Severity: High.
- Affected Systems: Corporate email, VPN, internal Applications.

⇒ Investigation Findings:

- multiple employee email addresses were exposed.
- some credentials matched active company accounts.
- The leaked credentials likely originated from a previous data breach.
- No confirmed unauthorized access was detected, but the exposure presented a significant risk.

⇒ Response Actions:

- Reset password for affected users.
- Force password changes.
- Enable multi-factor Authentications.
- monitor login activity.
- Block suspicious IP addresses.
- Notify affected employees.
- Investigation possible account misuse.

⇒ Impact Assessment:

- Potential unauthorized account access
- Risk of data theft.
- Possible privilege escalation.
- Increased phishing risks

⇒ Conclusion:

Early detection and prompt response prevented account compromise and strengthened security.